

WASTE → VALUE

CleanLoop Biogas

Sanitation → energy

Bangladesh's faecal-sludge gap meets viable, profitable biogas-and-fertilizer resource recovery



THE SITUATION IN BANGLADESH

In urban Bangladesh roughly 42% of residents depend on on-site sanitation, yet faecal sludge is handled mostly by uncoordinated, informal pit-emptying markets rather than formal treatment, leaving a major safety and treatment gap ¹. The cost of that gap is staggering: the World Bank's Water and Sanitation Program estimated inadequate sanitation cost Bangladesh Tk 295.48 billion in 2010, about 6.3% of GDP, with around 40% of people still relying on shared, rudimentary facilities ⁵. At the same time the country generates roughly 124 million tonnes of livestock and urban waste each year and could in principle support about 5 million biogas plants, but only around 150,000 have ever been built ⁶. The same waste streams that threaten public health are therefore a largely untapped energy and fertilizer resource sitting in plain sight ⁶.

OUR PITCH

CleanLoop Biogas runs a paid faecal-sludge-management service and pays for itself by selling what it recovers, because faecal sludge and organic waste can be converted by anaerobic digestion into biogas energy, soil-conditioning biofertilizer and feed ². The unit economics are proven, not hypothetical: a 2022 Heliyon study of 300 household biogas plants in Bangladesh measured a net annual benefit of about USD 295 per plant and a 43% internal rate of return even without subsidy ³, and decentralized recovery elsewhere has cut household energy costs by up to USD 2.07 per day and fertilizer costs by up to 72.6% ⁸. Our real wedge is operations: that same study showed plant survival hinges on follow-up maintenance service (p<0.001) and operator training (p=0.045), so CleanLoop builds the recurring service-and-training layer that informal markets lack ³, and we raise yields by co-digesting cow manure — slightly below the optimal 20–30 C:N ratio on its own — with higher-carbon waste to stabilize digestion ⁴. The deployment-and-finance model is already de-risked in-country: under the National Domestic Biogas and Manure Programme, IDCOL with SNV financed roughly 37,269 household digesters between 2006 and 2012 ⁷. We are honest that scaling FSM-fed digesters to community scale still needs field validation, but the core economics and the operations gap they expose are well-evidenced ²³.

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